Computer Graphics and Visualization

(Graphics APIs and Pipelines)

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Week 2

Learning Objectives

By the end of this lecture, students should be able to:

- see the main areas of computer graphics and the wide range of applications where it is used today.
- understand the concepts of graphics APIs and pipelines and explain their importance.

Applications of Computer Graphics

Computer Graphics covers basics of graphics, generating 2D and 3D images in computers using different algorithms and several exciting approaches. From video games and movies to medical imaging and simulations, computer graphics are used everywhere.

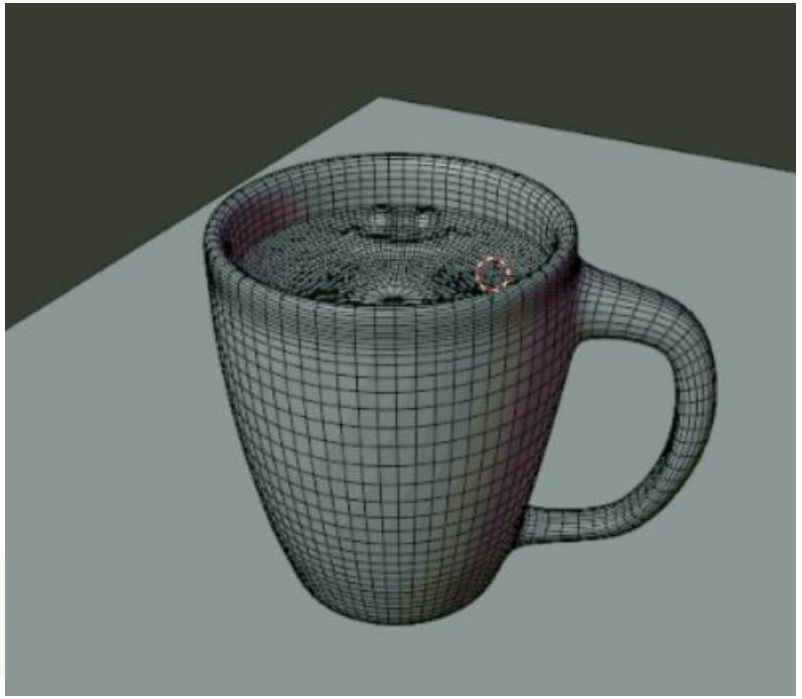
Major Areas of Computer Graphics

We can categorize Computer Graphics into several areas, each focusing on different aspects of image creation and manipulation. The core areas include modelling, rendering, and animation, but there are other related fields as well.

Modelling in Computer Graphics

- Formally, modelling is the mathematical description of shapes and their appearance. Modelling states defining objects in a way that computers can store and manipulate.

- For example, a coffee mug can be described as a set of ordered 3D points, connected by an interpolation rule, and combined with a reflection model to simulate how light interacts with the mug.



Major Areas of Computer Graphics *cont.*

Rendering in Computer Graphics

- Rendering refers to the process of generating shaded images from 3D computer models. It involves applying lighting, shading, and textures to the objects to create a final image.

- The term "rendering" comes from the world of art, where it refers to the technique of adding shading to drawings to make them look three-dimensional.

This image is not a photograph; it is 3D modelled with textures, shading, shadows, environment maps, materials etc. **This image has been created using Blender 3D**.

Major Areas of Computer Graphics *cont.*

Animation in Computer Graphics

- Animation adds movement to models by creating a sequence of images that simulate motion. While modelling and rendering focus on still images, animation brings those images to life by adding the dimension of time.

- For example, in a cartoon, an animated character walking across the screen is achieved by rendering multiple frames where the position of the character changes slightly between each frame.

The Role of User Interaction in Computer Graphics

- User interaction is another important aspect in computer graphics that bridges the gap between humans and computers. It is used to manage how users control and interact with digital environments through input devices like a mouse or keyboard.

- For example, in computer-aided design (CAD) software, designers use the mouse to rotate, zoom, and manipulate 3D models on the screen.

Virtual Reality (VR)

- Fairly a new concept is virtual reality. This immerses users in a computer-generated 3D environment.
- VR applications often involve stereo graphics and respond to head motion to create the feeling of being present in a virtual world.
- Advanced VR setups may also include sound and force feedback.

- As an example, in VR gaming, players wear a headset that lets them explore virtual worlds as if they were inside the game.

Visualization in Computer Graphics

- All the fields we are discussing can be used as visualization purposes. But this shows the process of transforming complex data into visual formats to help users understand it more easily.
- This can include graphs, charts, and other visual tools that make abstract information clearer.

- For example, a weather forecasting system that visualizes temperature changes over time using graphs is an example of data visualization.

Image Processing in Computer Graphics

- Graphics and Image processing are different but closely related. Image processing involves manipulating and transforming 2D images.
- This can include tasks like filtering, resizing, or enhancing images. Image processing is commonly used in both computer graphics and computer vision.

- One such common example in Image processing is removing noise from a photograph using software like Photoshop.

Graphics APIs and Pipelines

Graphics APIs and Pipelines

- Graphics APIs and pipelines are crucial in computer graphics. They enable developers to create visuals for video games, animations, or simulations. These systems are important for managing how images are drawn and displayed on the screen.

What are Graphics APIs?

- A Graphics API (Application Programming Interface) is a set of functions that allows programs to perform operations like drawing images and 3D surfaces. Essentially, APIs provide a way for software to communicate with hardware. Graphics APIs focus specifically on creating and displaying visual content.

- Every graphics program requires two key types of APIs:

- A Graphics API to handle the visual output
- A User-Interface API to manage user input

Types of Graphics APIs

There are two approaches for using graphics APIs –

Integrated Approach

- Some languages, such as Java, have built-in graphics and user-interface APIs. This ensures portability. Where the code will work across different systems. Everything is standardized, making development simpler for beginners.

Library-Based Approach

- Direct3D and OpenGL are example of these types, where drawing commands are part of a software library. This approach is more powerful but can vary from system to system. It offers more control, portability can be an issue unless developers use an additional library to handle user interface differences.

Examples of Popular Graphics APIs

Direct3D and OpenGL are the most popular and widely used Graphics APIs.

- **Direct3D** – It is used in video games, the Direct3D is a graphics API that belongs to the DirectX suite by Microsoft. It focuses on rendering 3D graphics, allowing games to render realistic environments. The popular game Halo uses Direct3D to render its 3D environments.

- **OpenGL** – It is a cross-platform graphics API that is widely used for developing both 2D and 3D graphics. It is particularly popular in CAD applications and scientific visualization. The graphics in Google Earth use OpenGL to render detailed maps and 3D landscapes.

What are Graphics Pipelines?

- After the graphics APIs, let us see the concept of graphics pipelines.
- These are the process that computers follow to create 3D images and display them on a 2D screen.
- It is a series of steps that take the input, such as the geometry of 3D models, and turn it into pixels on the screen.

Stages of a Graphics Pipeline

- Let us understand the steps for graphics pipeline –

- **Vertex Processing** – This is the first step where the positions of 3D points (vertices) are calculated. These vertices form the basic structure of 3D objects. For example, we are rendering a car in a racing game, the position of each vertex of the car model is calculated in this stage.
- **Clipping and Culling** – Once the vertices are processed, the system decides which parts of the scene are within the cameras view and which will not. The objects outside the view are discarded (culled). The clipping stage trims any part of an object that is partially outside the view. For example in a first-person shooter game, only the objects visible to the player's viewpoint are kept, while the rest are culled. Similarly far objects from camera are culled.

Stages of a Graphics Pipeline

- **Rasterization** – The next important step is rasterization. This is where the 3D models are converted into pixels or fragments. These pixels represent the final image that will appear on the screen. For example the 3D model of a tree in a game is converted into a collection of colored pixels that represent the tree on the screen.
- **Fragment Processing** – In this step, each pixel is assigned its final color and texture. The system applies lighting and shading to make the image more realistic. This is like when sunlight shines on the tree in a game, fragment processing ensures that the tree looks appropriately shaded and textured.
- **Output Merging** – Finally, the output merging. It takes the colored fragments and determines which pixels are visible. This process ensures that objects in the front are displayed while those behind are hidden. Like If a car drives behind a building, the car will be hidden from view because the output merging ensures the buildings pixels appear in front of the car's.

Graphics Pipelines in Hardware

- Graphics pipelines are closely related to hardware. Most computers today have a built-in 3D graphics pipeline in their hardware. These pipelines are optimized to draw 3D shapes and transform them into 2D images efficiently. This hardware usually focuses on processing 3D triangles, the basic shape used in most models.

- **The Role of Triangles in Graphics Pipelines** – In computer graphics we use the concept of triangles very often. This is because they are simple shapes that can easily be processed by computers. Most 3D models are made of thousands, if not millions, of small triangles.

Graphics Pipelines in Hardware

- **Z-Buffering** – This handles depth in graphics pipelines. In complex 3D scenes with many objects the correct objects are displayed in the right order is a challenge. The Z-buffer is a technique used in the pipeline to manage the depth of objects. Objects closer to the camera are drawn in front of those farther away.

- **Level of Detail (LOD) Optimization** – Another aspect is LOD optimization. More efficient systems often use Level of Detail (LOD) which means that objects far from the camera are rendered with fewer triangles, as less detail is needed. As the object gets closer, more triangles are used to improve the detail.

THANK YOU FOR LISTENING